

Cabe Robertson

caberobertson@gmail.com • 575-644-4703 • caberobertson.github.io

OBJECTIVE

Seeking a Summer/Fall 2026 internship or co-op in R&D, Test, or Hardware Engineering. Hands-on experience in embedded systems, RF and signal processing, PCB design, and hardware prototyping across multiple engineering internships and research.

EDUCATION

New Mexico State University — Las Cruces, NM

Expected December 2027

- B.S. in Electrical Engineering — Concentration in Communications & Signal Processing

TECHNICAL SKILLS

- **Programming & Software:** Python, C, C++, MATLAB, VHDL, Arduino IDE, OpenCV, Git/GitHub, LaTeX (Overleaf)
- **Hardware & Embedded:** PCB Design (EAGLE), FPGA, ESP32 & Microcontrollers, BLDC Motor Control (PWM), Soldering
- **Test & Lab Equipment:** Oscilloscope, Function Generator, Power Supply, Multimeter, Anemometer
- **Languages:** English (Native), Spanish (Native)

PROFESSIONAL EXPERIENCE

Hardware Engineering Intern — Syndetix Incorporated, Las Cruces, NM

Apr 2025 – Aug 2025

- Designed and built a portable \$219 demonstration kit for an RF communication-jamming device shown to law enforcement and defense clients, ~56% cheaper than the prior \$500 setup
- Engineered a lightweight stabilizing fix that kept the demo operational in windy outdoor conditions, eliminating tip-over failures during client demonstrations
- Resolved a 4-year-old server permission/synchronization bug, restoring reliable shared-file access and improving workstation efficiency company-wide
- Authored the product user manual, standardizing setup and handling procedures for mission and training use

Electrical Engineering Intern — GILZ LLC, Remote

Mar 2026 – May 2026

- Reverse-engineered a desk blower fan to characterize its BLDC motor's voltage/current ratings and PWM response for an air-filtration system prototype
- Programmed an ESP32 (Arduino) for 3 user-selectable fan speeds, adding speed control where the stock unit had none
- Benchmarked airflow, noise, temperature, and velocity across duct and filter attachments, identifying the configuration that reduced noise by ~10 dB and increased airflow by 20% versus the stock fan

Research Assistant — Mitchell Coding Group, Las Cruces, NM

Aug 2025 – Feb 2026

- Developed Python simulations of linear-algebra channel decoding for wireless models, validating bit-error performance across various encoding schemes
- Designed and ran multi-hour experiments over contrasting input bit-string values to verify decoder reliability under varied conditions
- Automated result processing and plotting in MATLAB, producing publication-quality figures

ACADEMIC PROJECTS

Face Tracking System — Python (OpenCV), Arduino, Pan-Tilt Servos

- Built a real-time computer-vision system that detects faces and streams positional data over serial to an Arduino, driving pan-tilt servos to keep the subject centered at 30 FPS

Self-Navigating Robot — Arduino, C++, Ultrasonic Sensors

- Programmed an autonomous obstacle-avoidance robot with ultrasonic sensors, refining control logic over repeated trials to achieve a 100% obstacle-avoidance rate in testing

Seven-Segment Display — VHDL, FPGA

- Designed a VHDL display driver decoding all 16 BCD/HEX inputs (0–9, A–F) on an FPGA; optimized the Boolean logic with truth tables and Karnaugh maps to minimize gate count

LEADERSHIP

Vice President — IEEE NMSU Student Chapter

Oct 2024 – Present

- Lead a \$50,000 fundraising initiative to revitalize the chapter and ECE department
- Spearheaded the chapter's turnaround, earning its Most Improved Student Organization Award
- Grew general-meeting attendance to 30+ students and mentor members in hands-on troubleshooting and soldering workshops
- Serve as primary liaison to industry sponsors, national IEEE, and campus organizations to secure collaborations and event sponsorship